FIT3159 COMPUTER ARCHITECTURE

UNIT INTRODUCTION

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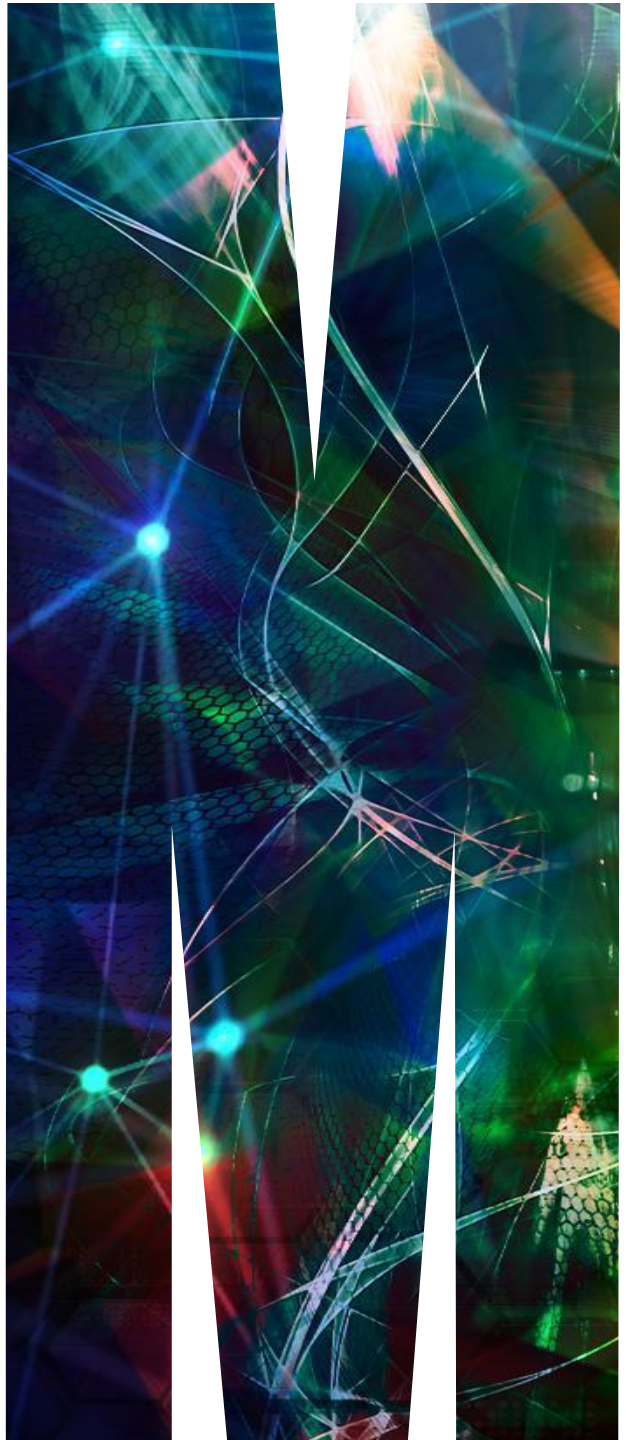


ABOUT THE UNIT

A LOOK AT THE DEEP FUNDAMENTALS

The internal mechanism of computers and how they are organised and programmed.

UNIT STRUCTURE

LECTURE TOPICS

- Combinatorial and sequential logic
- Memory addressing and data representation
- Caches and cache architectures
- Pipelined architecture and superscalar architecture
- RISC and CISC machine architectures

THE TEACHING TEAM

- **Lecturers (Clayton Campus)**
  - Dr. Chetan Arora - Chetan.arora@monash.edu (Chief Examiner)
  - Dr. Charith Jayasekara - Charith.Jayasekara@monash.edu

- **Lecturer (Malaysian Campus)**
  - Dr. Adamu Muhammad Buhari - adamumuhammad.buhari@monash.edu

- **Tutors**

Details of the teaching staff can be found on Moodle

Moodle page -> Learning ->Additional Information and Resources -> Unit Teaching Team

OBJECTIVES

AFTER COMPLETING THIS UNIT, STUDENTS SHOULD BE ABLE TO

- Provide the student with an understanding of the fundamental design features and operating principles of a modern digital computer.
- Provide the student with the ability to understand the implications of various design strategies upon hardware and computer system performance.
- Provide the student with the ability to understand the relationship between the architecture of the machine and its instruction set.
- Provide the student with basic programming skills in assembly (machine) language.

ASSESSMENTS

IN-SEMESTER

- **Bi-Weekly applied classes (20%)**
  - During week 1,3,5,7,9 and 11
  - 2 hours long
  - Flipped (except for week 1)

- **Bi-Weekly laboratory classes (30%)**
  - During week 2,4,6,8,10 and 12.
  - 3 hours long

- **Applied Problem Solving Tasks (50 %)**
  - Problem Solving Assessment 1 - Week 5
  - Problem Solving Assessment 2 - Week 9
  - Problem Solving Assessment 3 - Week 12

NO FINAL EXAM

TEACHING ACTIVITIES

- **PRE-RECORDED LECTURES**
  - Will be available under weekly content in Moodle
  - Ensure to view these videos prior to attending the classes.

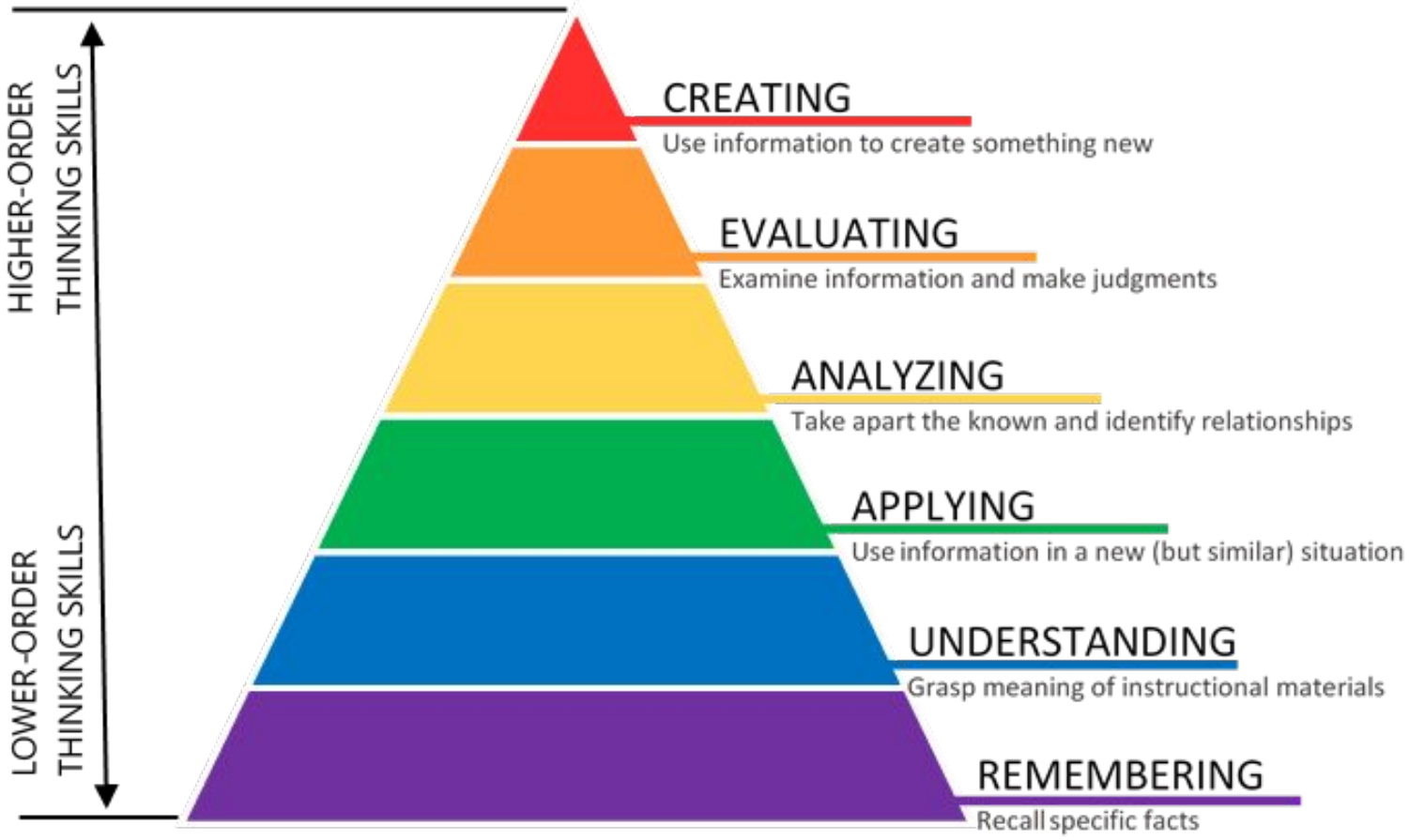
- **WEEKLY WORKSHOPS**
  - 2 Hour sessions.
  - Start from Week 1

- **BI-WEEKLY TUTORIALS**
  - 2 hour sessions
  - Will be conducted during odd weeks of the semester
  - Start from Week 1

- **BI-WEEKLY LABS**
  - 3 Hour sessions.
  - Will be conducted during even weeks of the semester
  - Start from Week 2

FACING LEARNING CHALLENGES

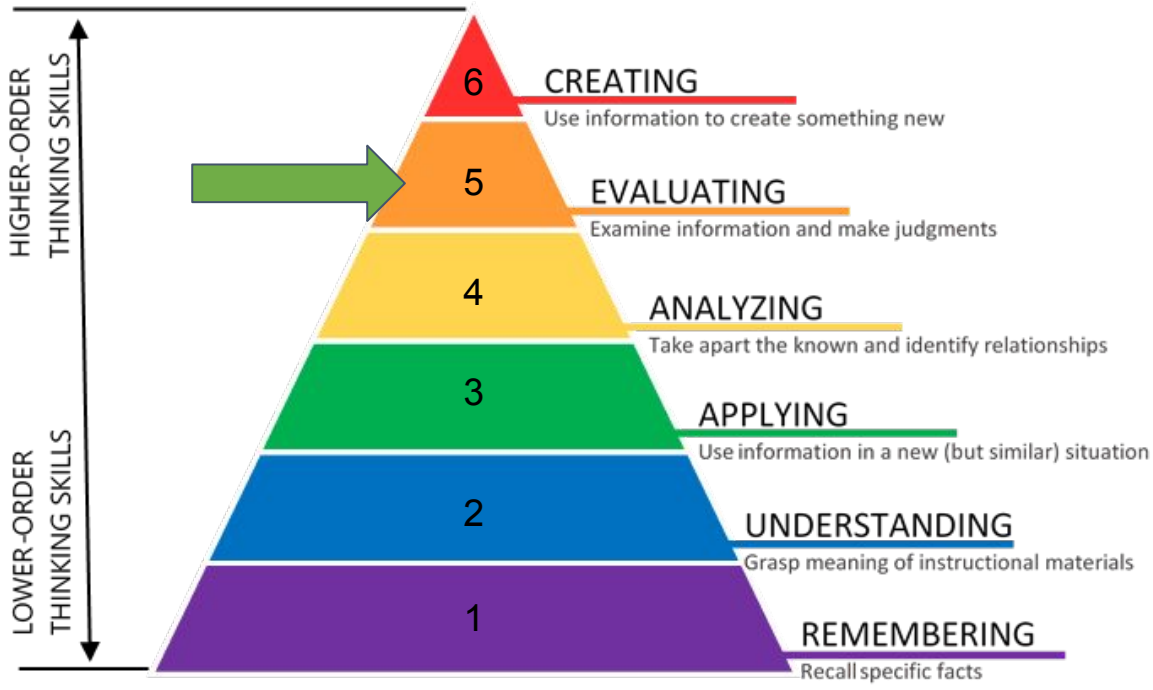
LEVELS OF LEARNING - BLOOM'S TAXONOMY



LEVELS OF LEARNING

THIS UNIT GOES UP TO LEVEL 5

- FIT3159 is usually taken at the beginning of third year.
- Intermediate level of learning.
- **Being able to compare and link different concepts together to solve non-trivial problems.**
- Bad way to learn: memorise every possible answer
- Good way to learn: attempt to generalize and link concepts so you can deduce answers based on your own understanding



DUNNING-KRUGER EFFECT

A COMMON MENTAL BARRIER TO MORE ADVANCED LEARNING

- People who think they are experts tend not to be.
- People who are experts tend to underestimate their knowledge.
- As we learn, we create a mental model of the knowledge we gain. We can only gauge our own expertise in terms of our own mental model.
- When your own mental model is full, it feels like there is nothing more to be learned: more difficult to accept new information that does not fit a preconceived model. Feels like ‘hitting a wall’ in learning progress, also feels like there is nothing more to be learned, and further knowledge is ‘wrong’.
- Everyone hits this wall eventually.
- Sometimes old assumptions need to be questioned or broken before new information can be learned or utilised.

RESOURCES

PRESCRIBED TEXTBOOKS

- Stallings 8th Ed.
- Mano & Kime 4th Ed.
- Tanenbaum A S Structured computer organisation 3rd edn, Prentice-Hall, 1990

- Or any good book on computer architecture.

Access to these books are available online via the library.

LINUX VIRTUAL MACHINE

COMMON PROGRAMMING ENVIRONMENT FOR FIT3159 / FIT3142 and FIT2100

For the first half of the semester, VMs aren't required as our lab software is compatible with Windows, Mac, and Linux. However, we're preparing VMs for later use:

- **VM Availability:** Details will be posted on the Moodle unit information page.
- **Usage:** VMs will run on VirtualBox (free) on your laptop, essential for specific tasks like Week 12's Lab 6.
- **Alternatives:** For those unable to use this VM (because this may not run on apple silicon), a cloud-hosted option will be available soon on Moodle.

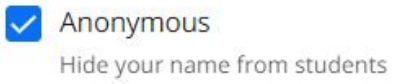
UNIT INFORMATION / LEARNING PAGES

GO TO MOODLE -> UNIT INFORMATION or LEARNING

- Virtual machine download links
  - Try to get your Linux environment working as early as possible when the information is available
- Free eBooks (reproduced with permission from author)
  - These don’t cover the whole course, but some are useful references for the first assignment
- Contact details for relevant teaching staff

FORUMS AND ANNOUNCEMENTS

- Important announcements may be made through Moodle
  - You should get an email notification when this happens
- Ed forums for engaging with topics/concepts outside classes
  - For engaging with the content of the course
  - Disruptive posts are inappropriate here
    - e.g. Posting to undermine an assessment task instead of trying to understand or overcome an unfamiliar challenge



<- Note, this hides your name from *students* only

ACADEMIC INTEGRITY, PLAGIARISM AND COLLUSION

YOUR RESPONSIBILITY

- Study ethically and with honesty and integrity
- Appropriately acknowledge the work of others
- Study independently on individual assessments
- Make sure academic work is not falsified or altered fraudulently
- Take reasonable steps to make sure other students can't copy or misuse your work.
- Penalties include:
  - a zero mark for the assessment task
  - a zero mark for the unit
  - suspension from the course
  - exclusion from the University.

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Summary

YOUR TO-DO LIST…

- Try to prepare for Workshops, Applied and Labs before the class, after material is released
- Get familiar with the reading materials on the Unit Information and Learning pages, you might need them later in semester